

PCS–CTL control board

Rev A0 | 2026–10–05 | Schematics authored by Chinmoy Bhuyan | PCS–P125 inverter controller, PV–CTL assembly variant: F280039C on DC–, one trip latch

01 PC connector, supply, reset

2x32 PC contract to the power board (all LIVE): only +3V3 is used (200 mA allocated by PV–PWR); TPS3828 reset + 1.6 s watchdog

File: sheets/01_supply.kicad_sch

02 F280039C, clock, debug, EEPROM

TMS320F280039C (100–pin PZ), decoupling, 25 MHz SPXO, cTI–20 JTAG (LIVE), EEPROM, boot straps, command pull–downs, run LED

File: sheets/02_mcu.kicad_sch

03 Reference and analog front ends

REF3030E 3.0 V = VREFHI = VDAC = NTC bias = trip thresholds; IL1–4 gain stages; power–board signals into the ADC through 100 R / 1 nF; 8 NTC biases + inlet NTC

File: sheets/03_afe.kicad_sch

04 Hardware trip comparators

7 x TLV9024 on +3V3 (open drain, no firmware): IL windows from each sensor's Uref; VA / VB over–voltage, NTC over–temperature and open probe from VREF ladders

File: sheets/04_trip.kicad_sch

05 Trip latch, watchdog, gating

74LVC07 wired–AND of every trip source → PRE_N → 74LVC1G74 latch (no firmware); heartbeat monoflop; 74LVC08 gating of 16 PWM, EN, K_A, K_B, K_PRE, STATUS, FAN

File: sheets/05_latch.kicad_sch

06 Barrier B1: isolators, CAN, RS–485

Reinforced barrier B1: CA–IS3050W CAN, CA–IS3082WNX RS–485, 2 x CA–IS3821LG, CA–IS3820LG; bus TVS, jumper terminations, pluggable field terminals (SELV)

File: sheets/06_iso.kicad_sch

07 SELV supply, fans, enable / status

SELV 24 V input (AUX–T1 reinforced winding, 12–36 V), TPS54360B 5 V and fan supply, 3 fan headers + tach, ENABLE / STATUS terminal, SELV 0 V to PE

File: sheets/07_selv.kicad_sch

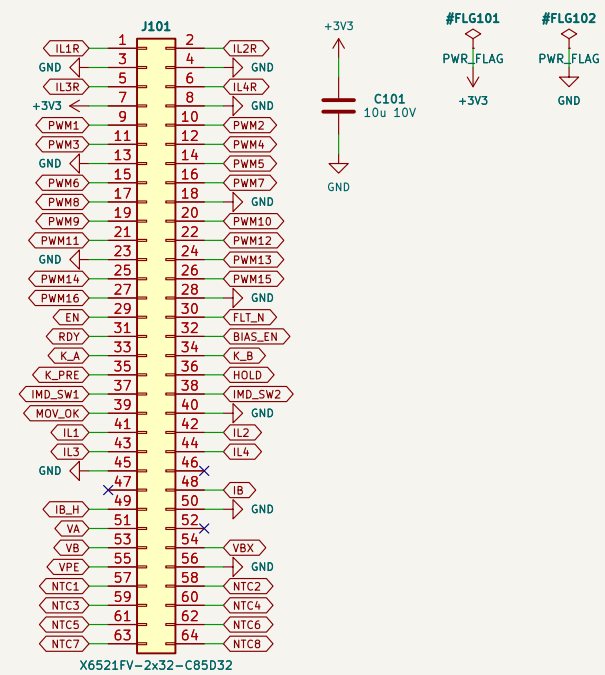
08 PCS_X: grid voltages, RCM, NTC9, K_AC2

2x8 PCS_X contract to PCS–PWR (all LIVE): K_AC2 from the gating (sheet 05), RCM_TST from GPIO21; grid voltages, RCM and NTC9 into the ADC

File: sheets/08_pcsx.kicad_sch

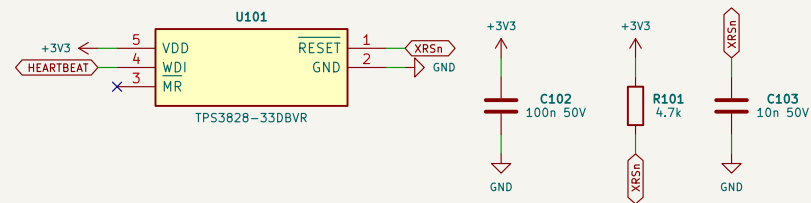
ARCHITECTURE–PCS section 6; the PV–CTL schematic re–valued by gen/pcs_ctrl.py; PCS_PC / PCS_X levels as built on PCS–PWR (hardware/PCS–PWR/outputs/PCS–PWR_design_check.txt).
Assembly option: the two TLV9024 marked ‘four–wire only’ are not fitted on the three–wire build (bom/PCS–CTL–3W_BOM.csv).
Pin plan: gen/data/pcs_ctrL_pin_plan.csv (from SPRSP61C, checked every build).

PC CONNECTOR (GEN/INTERFACES.PY PC; AGND = GND ON THIS BOARD)



LIVE: GND = AGND = BUS-; nothing on this connector leaves the enclosure. Only +3V3 comes across (the former +24V / +5V pins carry the sensor references IL1R-IL4R since PV-PWR rev A2). +3V3 feeds the MCU, so the power board's 3.3 V signals and VDDIO share one rail and one power-up order (no back-powering).

RESET AND WATCHDOG (TPS3828-33)



RESET (open drain) holds XRSn below 2.93 V and for 200 ms after; no WDI edge for 0.9-2.5 s -> reset. XRSn 4.7k pull-up and 10 nF (SPRSP61C table 5-1: 2.2-10k, <= 100 nF). XRSn also clears the heartbeat monoflop, so every MCU reset sets the trip latch. +5V / +24V supervision: on the power board (RDY).

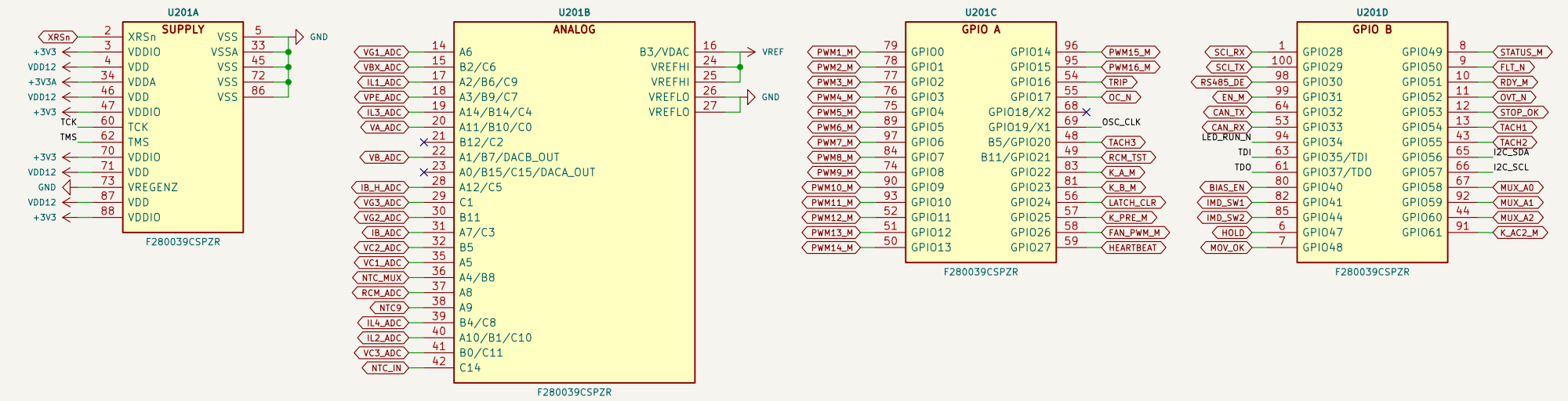
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 2
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /01 PC connector, supply, reset/
File: 01_supply.kicad_sch

Title: PCS-CTL control board - PC connector, supply, reset

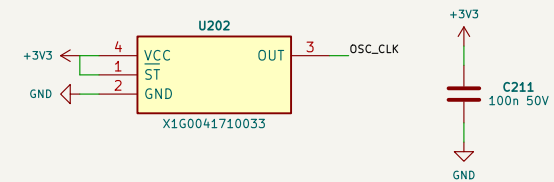
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 2/9

F280039CSPZR



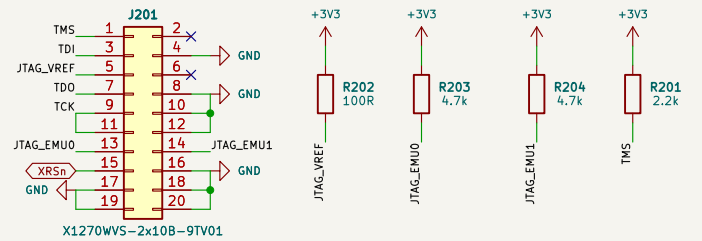
Pins as gen/data/pv_ctrl_pin_plan.csv (checked against SPRSP61C table 5-1 on every build). VREGENZ = GND; internal 1.2 V VREG; VDD only decoupled.

25 MHZ CLOCK (X1 SINGLE-ENDED)



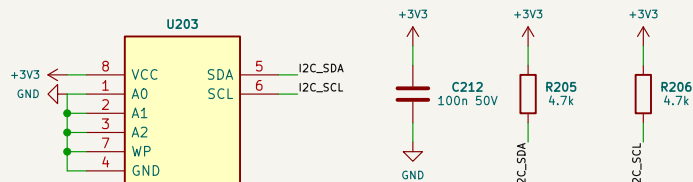
SG-210STF ± 50 ppm (CAN bit timing; CAN boot needs the external clock).
f(X1) 10-25 MHz, tr/tf ≤ 6 ns (SPRSP61C 6.12.3.2.1).

JTAG CTI-20 – HAZARDOUS LIVE (ISOLATED PROBE ONLY)



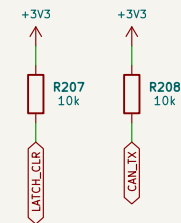
Label: 'HAZARDOUS VOLTAGE – up to 1000 V DC to earth – isolated debug probe only'. TMS 2.2k pull-up (no TRSTn on F28003x, table 5-1); TRST (2) open; EMU0/1 4.7k pull-ups; RESET (15) on XRSn; 17-19 GND. Field firmware update over CAN (DCAN boot option 1 or application loader); no live access needed.

EEPROM (CALIBRATION, SERIAL NUMBER, EVENT LOG)



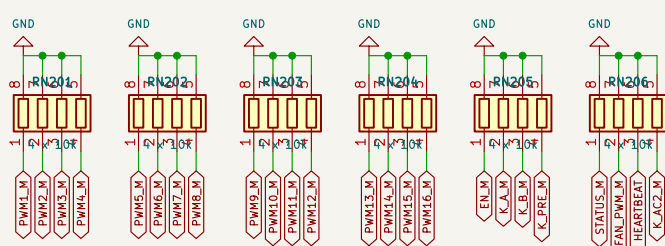
BL24C256A at 0x50 (A2-A0 low), WP low (writable; firmware guards the calibration pages). I2C 4.7k pull-ups.

BOOT STRAPS



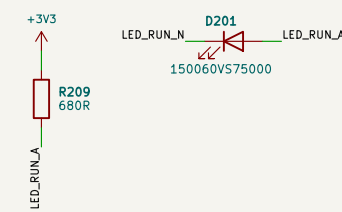
Flash boot = GPIO24 high and GPIO32 high at reset (SPRSP61C table 7-8); 10k pull-ups on LATCH_CLR (GPIO24) and CAN_TX (GPIO32, also recessive while the MCU is in reset).

COMMAND PULL-DOWNS (6 X 4 X 10K ARRAYS)



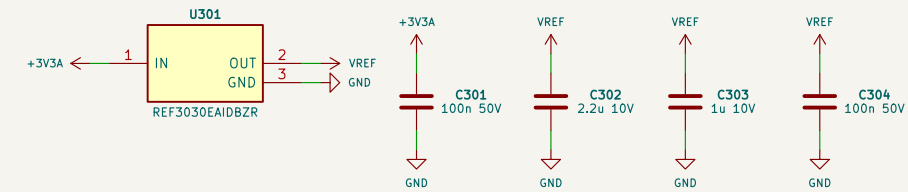
Every MCU output that ends in an AND gate here is low while the MCU is in reset or unprogrammed (GPIOs high impedance): 16_PWM, EN, K_A, K_B, K_PRE, STATUS, FAN_PWM, HEARTBEAT, K_AC2 (24 lines, 0 spare elements). BIAS_EN, IMD_SW1/2 go straight to PC: pulled down on the power board (10k / 100k, PV-PWR).

RUN LED



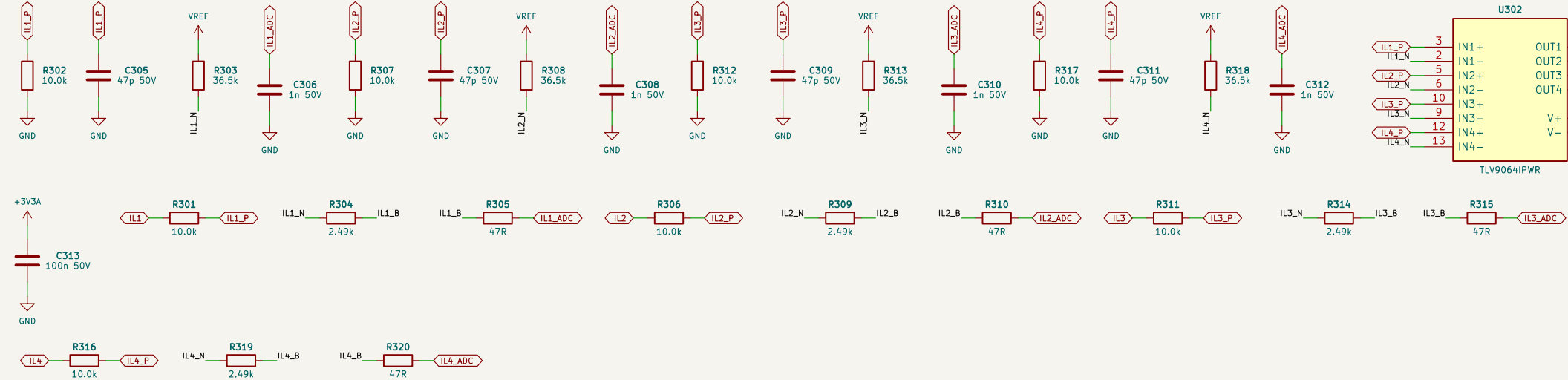
Green, GPIO34 low = on (firmware heartbeat), 680 R: ~2 mA.

3.0 V REFERENCE (REF3030E FROM +3V3A)



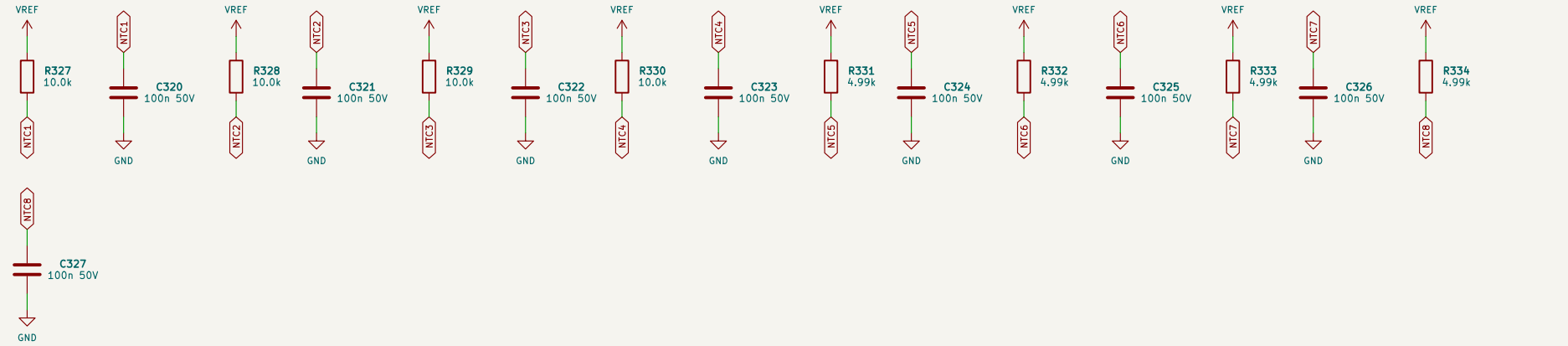
Feeds VREFHI (pins 24/25, >= 2.2 uF), VDAC (pin 16, >= 1 uF), the NTC biases, the IL offset and the threshold ladders of sheet 04: ADC, CMPSS and hardware trips are ratiometric to one reference. It runs from VDCA, so it never sits above VDCA (SPRSP61C 6.12.1.4.2). Needs +3V3 >= 3.20 V (REF30E dropout).

INDUCTOR-CURRENT FRONT ENDS IL1-IL4 (TLV9064 GAIN STAGES)



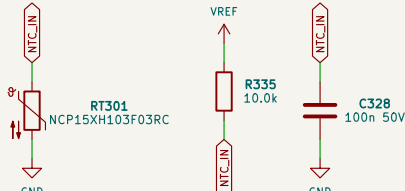
ILn = 2.5 V + 10.667 mV/A (STK-HO/A 75 on PV-PWR, 0.5-4.5 V, not ratiometric) -> 10.0k / 29.4k 0.1 %
-> ILn_P (47 pF, a = 0.740); trip window (sheet 04) and the + Input: Rg 36.5k to VREF; Rf 10.0k; ADC = 1.555 V + 10.14 mV/A (0.072 A/LSB, -153...+142 A) -> 47 R / 1 nF -> ADC + CMPSS; IL4 = 0 V on PV-P75.

NTC INPUTS (10 K B25/100 3988 K TO AGND + 100 NF ON PV-PWR, BIASED HERE FROM VREF)



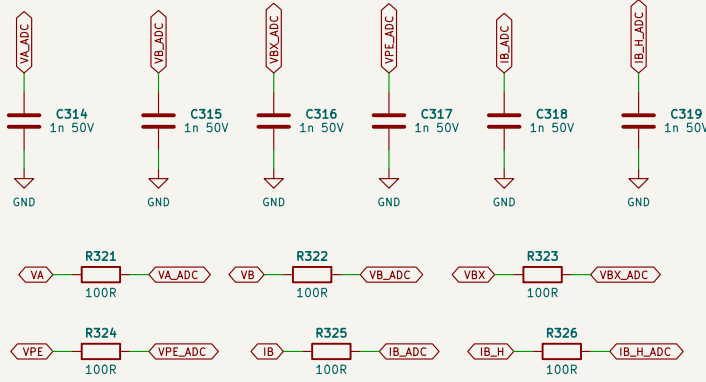
NTC1-4 heatsink sections (10.0k). NTC5-8 inductor windings (4.99k). 100 nF each. Open probe = 3.0 V; trips through the open-probe comparators (sheet 04); short = 0 V; trips as over-temperature.

INLET-AIR NTC (ON BOARD)



Murata NCP15XH103F03RC, 10.0k from VREF; ADC only (fan law, derating).

POWER-BOARD ANALOG SIGNALS (VMID 1.641-1.651 V PER PORT, NOT ON PC)



VA/VB/VAX/VBX/VPE = VMID + V/1203; IA/IB = VMID + 3.0 mV/A (+/-478 A); IA_H/IB_H = VMID + 1.0 mV/A. Anti-alias poles are on the power board; here 100 R / 1 nF COG as the ADC charge bucket. The trip comparators take the PC net itself (no ADC kick-back on them).

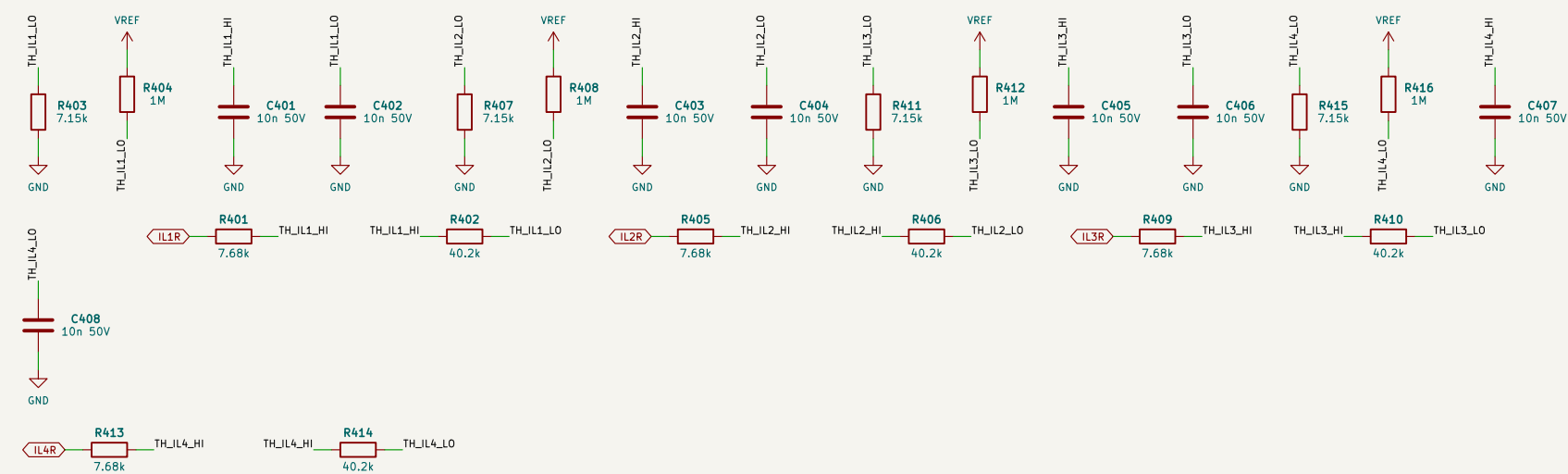
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 4
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /03 Reference and analog front ends/
File: 03_afe.kicad_sch

Title: PCS-CTL control board - Reference and analog front ends

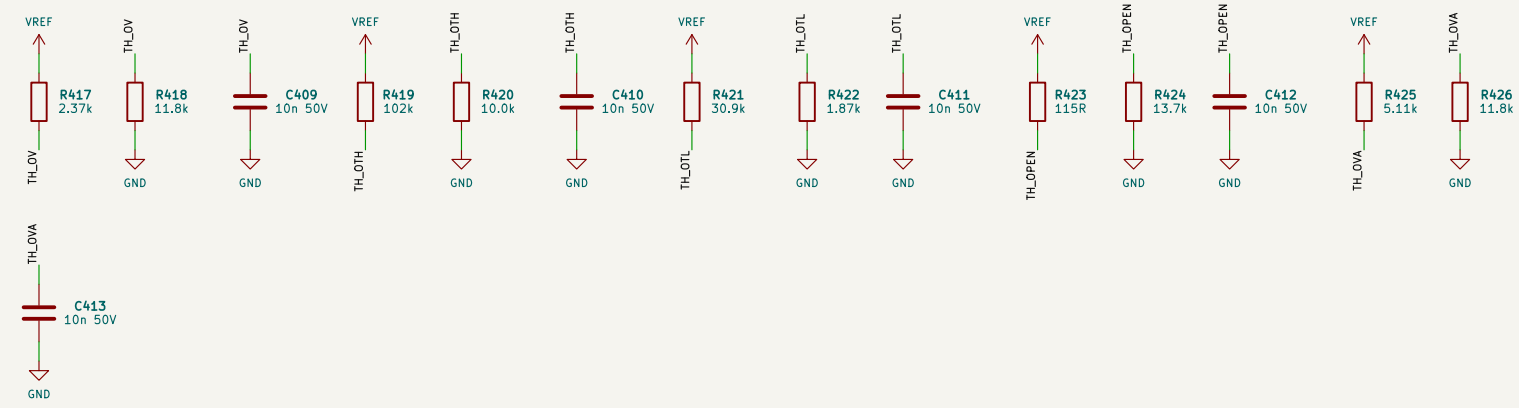
Size: A2 Date: 2026-10-05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 4/9

IL THRESHOLD LADDERS (0.1 %, 25 PPM/K), ONE PER PHASE FROM THAT SENSOR'S UREF (ILNR ON PC)



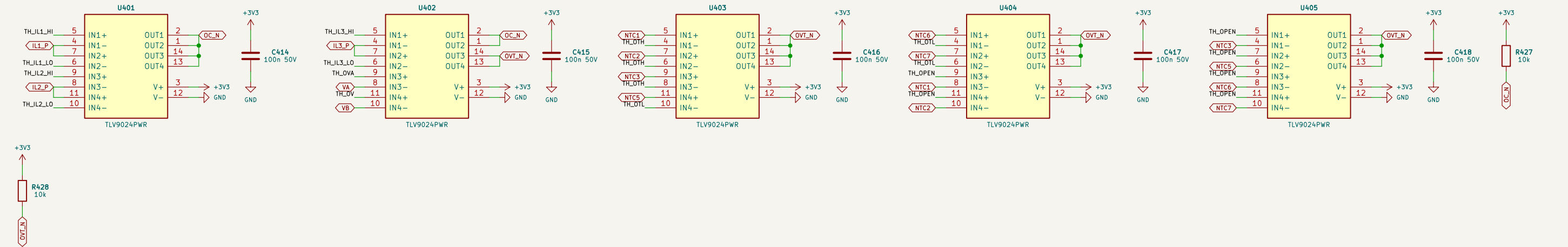
ILnR = 715R - TH_ILn_HI - 19.1k - TH_ILn_LO - 20.0k - GND: the window follows the sensor's own zero (Vref 2.48-2.52 V is no longer an error term; Voe +/-10 mV is), 1 M from VREF to TH_ILn_LO; with the sensor dead (IL = ILR = 0 V) TH_ILn_LO stays ~30 mV above ILn_P and the phase trips. Phase 4: fitted on every build.

THRESHOLD LADDERS (0.1 %, 25 PPM/K, FROM VREF)



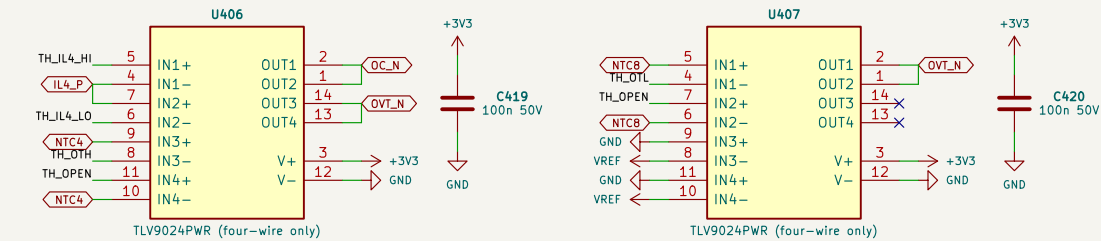
TH_OV: VA/VB 1084 V; TH_OTH: heatsink 87.9 C (PV-PWR: <= 89.9 C); TH_OTL: inductor 150 C;
TH_OPEN 2.975 V; an NTC node above it is an open probe (cold limit -30 C)

PHASES 1–3 AND THE PORT VOLTAGES (FITTED ON EVERY BUILD)



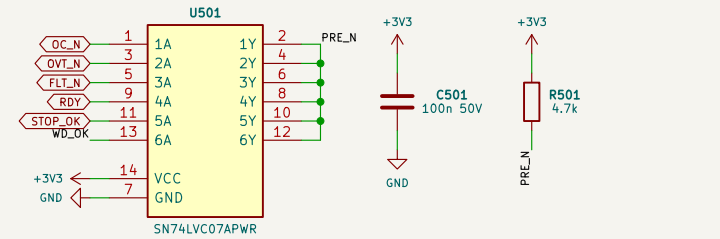
OC_N (wired-OR, 10k to +3V3); ILn_P above TH_ILn_HI or below TH_ILn_LO, n = 1-3. OVT_N (10k); VA, VB above TH_OV; NTC1-3 (heatsink) below TH_OTH, NTC5-7 (inductor) below TH_OTL; any of them above TH_OPEN.
Port over-current (387-413 A) arrives on FLT_N from the power board's own comparators (PW-PWR).

PHASE 4 – ASSEMBLY OPTION: FITTED ON PV-P100/110 ONLY



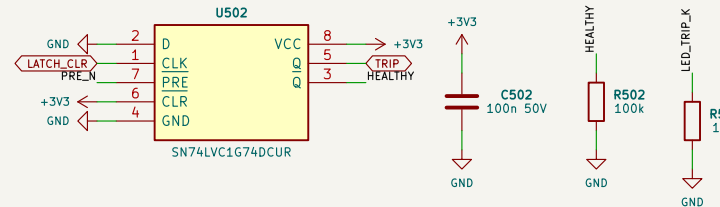
IL4 window, NTC4 / NTC8 over-temperature and open probe. Not fitted on PV-P75 (bom/PV-CTL-P75_BOM.csv) where IL4 = 0 V and NTC4 / NTC8 are open: nothing then pulls OC_N / OVT_N for phase 4. Fitted on a 4-phase build, a dead IL4 sensor (0 V) or an open NTC4 / NTC8 trips (fail-safe). Firmware cross-checks.

TRIP SOURCES --> PRE_N (OPEN-DRAIN WIRED-AND, 4.7K TO +3V3)



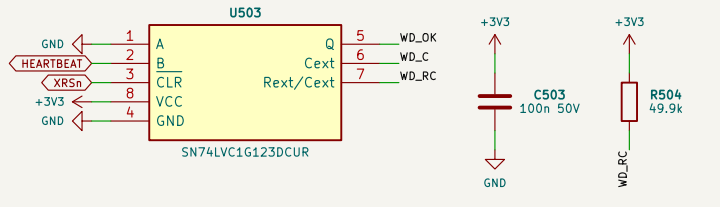
74LVC07 (loff, 5 V tolerant): DC_N, OVT_N, FLT_N (driver DESAT/UVLO + port OC of PV-PWR), RBY (gate supplies, +5V, live 24 V on PV-PWR), STOP_OK (ENABLE), WD_OK (heartbeat and MCU reset). Any low -> PRE_N low -> latch set asynchronously (Q = TRIP = 1).

THE LATCH (74LVC1G74, SET-DOMINANT)



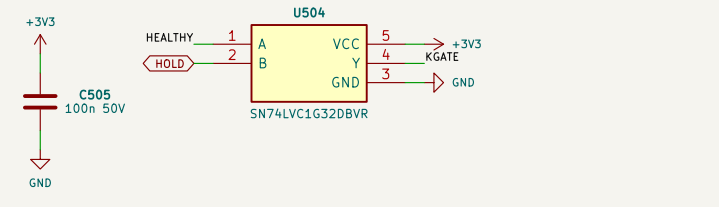
PRE = PRE_N, CLR tied high, D = GND, CLK = LATCH_CLR (GPIO24). A firmware rising edge loads 0 (clears) only while PRE_N is high (no source active); with any source active PRE wins (Table 9-1). Q = TRIP (red LED, INPUT/SMARTC -> T23), Q-bar = HEALTHY (gate). Unpowered: loff -> HEALTHY reads 0 (100%).

HEARTBEAT WATCHDOG (74LVC1G123)



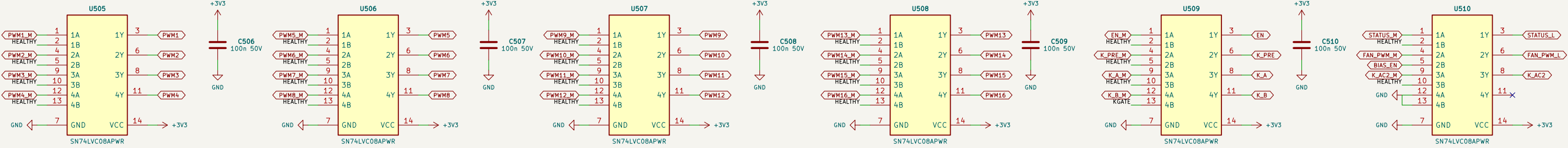
B = HEARTBEAT (rising edges from the control ISR, >= 500 Hz). A = GND, CLR = XRSn, Q = WD_OK stays high while edges come faster than tw = 49.9k x 100n (4.7-5.8 ms); no edge: a stuck pin or any MCU reset (XRSn low) -> WD_OK low -> latch. Rext to VCC, Cext between Cext and Rext/Cext (SCE586E table 4-1).

CONTACTOR GATE: KGATE = HEALTHY OR HOLD (74LVC1G32)



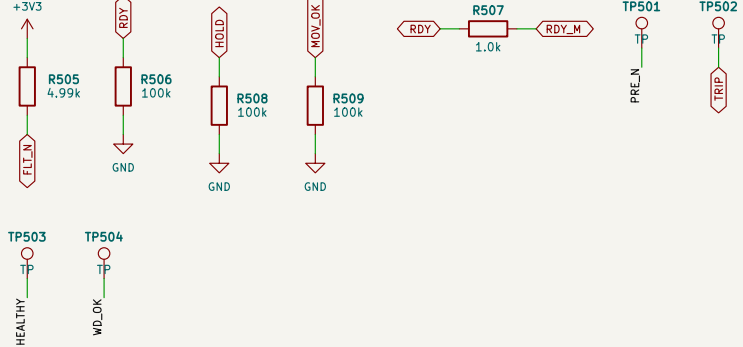
A trip drops K_PRE at once, but K_A / K_B only while HOLD is low; while the power board holds a contactor (port current above its breaking limit) this board never commands it open; the power board's coil interlock enforces the same. Closing still needs the MCU command AND (HEALTHY OR HOLD).

GATING: 6 X 74LVC08 (23 GATES USED, 1 SPARE WITH INPUTS GROUNDED)



PWm0 = PWM0_M AND HEALTHY (16), EN = EN_M AND HEALTHY, K_PRE = K_PRE_M AND HEALTHY, K_A/K_B = K_x_M AND KGATE, STATUS_L = STATUS_M AND HEALTHY, FAN_PWM_L = FAN_PWM_M AND BIASEN (PV-PWR rule: fans only while the gate bias loads the aux supply). The power board pulls every PC Input low (10k / 100k). PCS-CTL: K_A (AC contactor 3) and K_AC2 (AC contactor 2) = K_x_M AND HEALTHY; only K_B (DC contactor) keeps KGATE (HEALTHY OR HOLD).

POWER-BOARD STATUS LINES (PV-PWR AS BUILT)



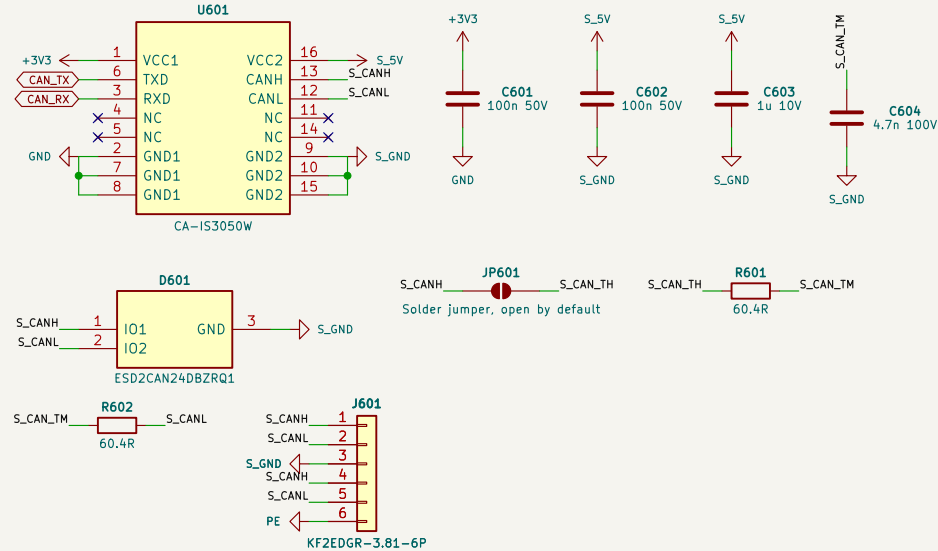
FLT_N: open drain on PV-PWR, pulled up HERE (4.99k, PV-PWR recommendation). RBY: open drain pulled up on PV-PWR (no pull-up here, 100k pull-down allowed), to the MCU via 1k. HOLD: push-pull. MCU_OK: open drain with 10k on PV-PWR; 100k pull-downs define an unplugged cable (not held / monitor open).

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 6
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

BARRIER B1 RULES (INSULATION_SPEC 'REINFORCED HV-PELV, DC POLE')

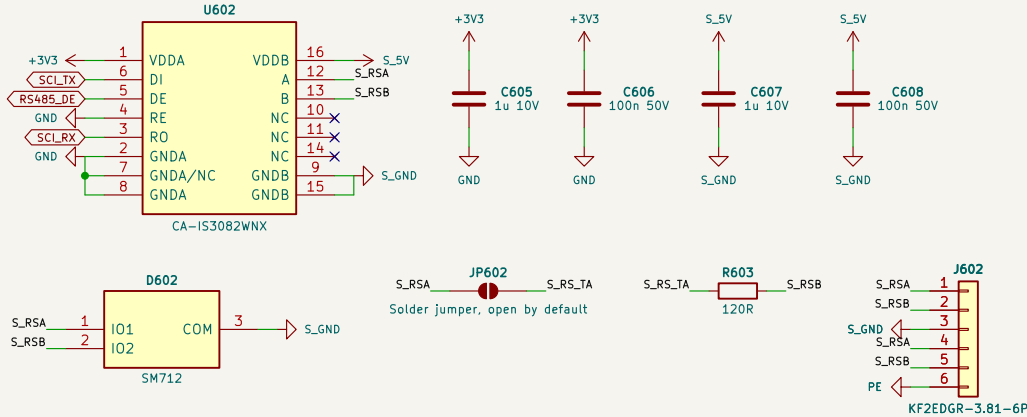
Every part on this sheet that touches LIVE and SELV is a declared isolator: VIOSM >= 8 kV, VISO >= 5 kVrms, VIOWM >= 1414 V DC, CLR/CPG 8 mm, 8 mm < 10 mm (PD2 creepage) -> conformal coating PD1 over the B1 zone is MANDATORY; routed slot under every isolator; SELV copper >= 10 mm from LIVE copper elsewhere.

ISOLATED CAN (BMS / PCS), CA-IS3050W



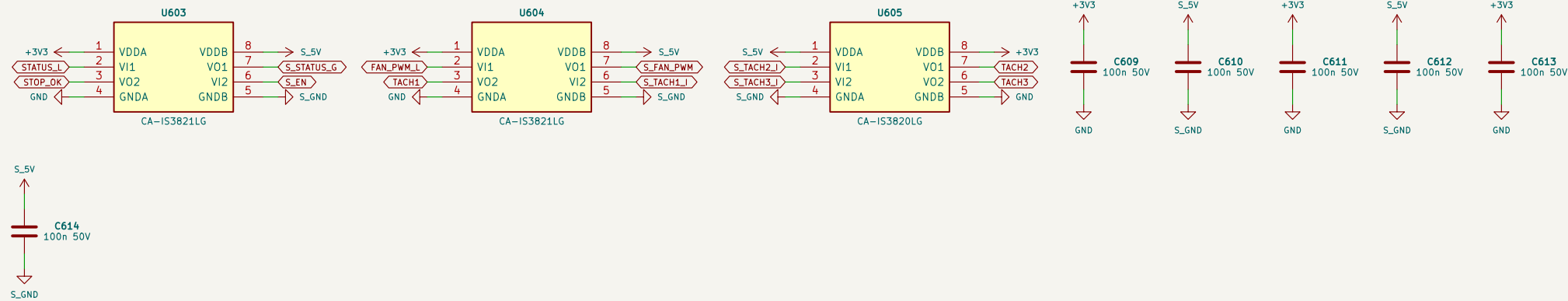
Logic side +3V3 (ICC1 <= 3.6 mA), bus side S_5V (ICC2 <= 73 mA dominant); no live +5V needed, Split termination 2 x 60.4 R + 4.7 nF by JP; ESD2CAN24 to SELV 0 V. Terminal: CANH, CANL, 0 V, CANH, CANL (in / out), shield to PE.

ISOLATED RS-485, CA-IS3082WNX



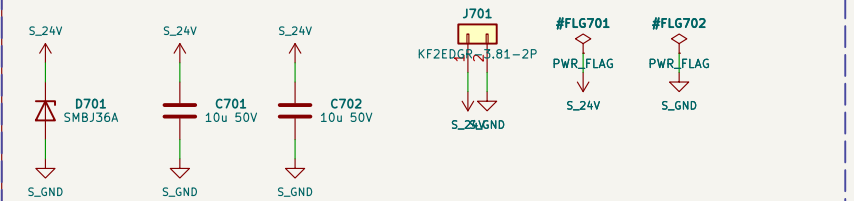
Logic side +3V3; bus side S_5V. DE from the MCU (internal pull-down; driver off in reset), RE tied low (receiver always on, fail-safe), 120 R by JP; SM712 to SELV 0 V. Terminal: A, B, 0 V twice, shield.

DIGITAL ISOLATORS (DEFAULT LOW = SAFE)



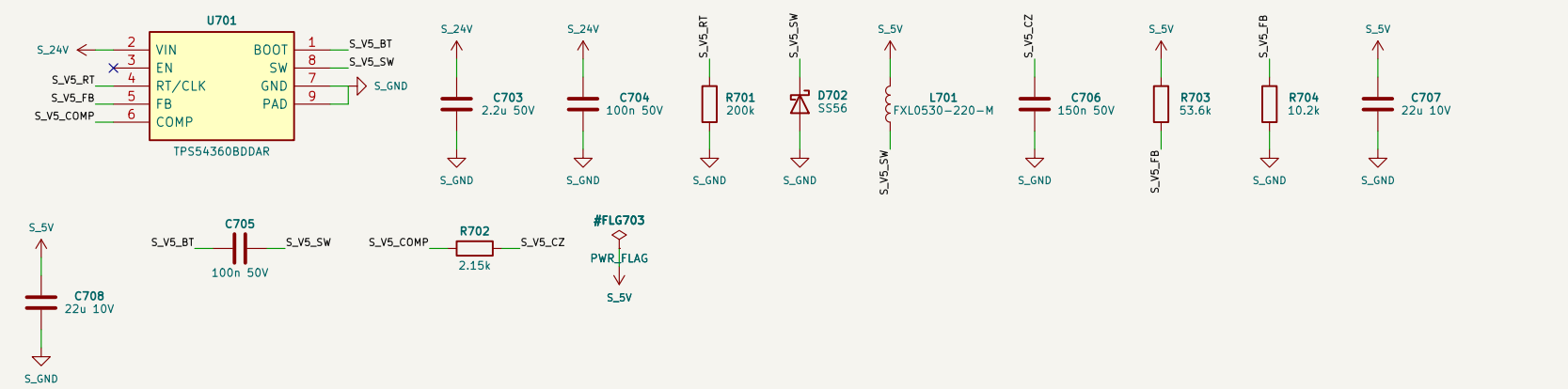
3821 #1: STATUS_L -> S_STATUS_G (out), S_EN -> STOP_OK (in; SELV side unpowered or open = STOP).
3821 #2: FAN_PWM_L -> S_FAN_PWM (out; unpowered = fans off), S_TACH1_I -> TACH1. 3820 (A side SELV):
S_TACH2_I, S_TACH3_I -> TACH2, TACH3, 100 nF per side.

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 7
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

SELV INPUT (2-PIN, DOUBLE-INSULATED LEADS, ≥ 10 MM FROM LIVE COPPER)

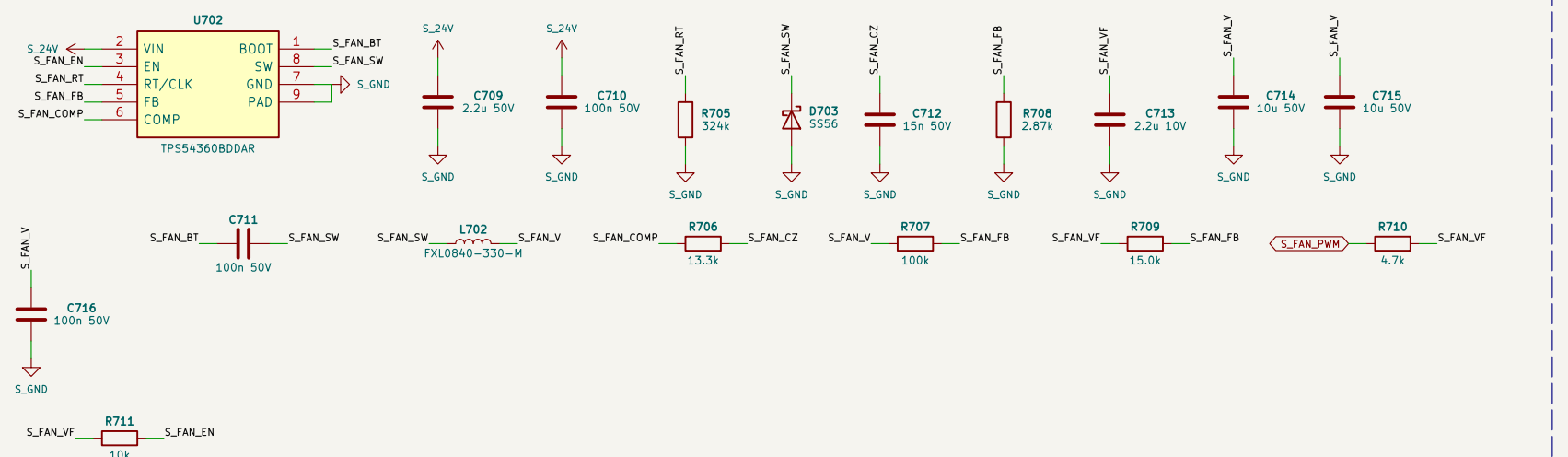
AUX-T1 fan winding, cross-regulated: 12-36 V (sags to -15 V without gate-bias load, PV-PWR). SMBJ36A clamp (36 V working). $2 \times 10 \mu\text{F}$ 50 V. Both bucks have no under-voltage lock-out above 4.5 V.

SELV 5 V (TPS54360B, 500 KHZ)



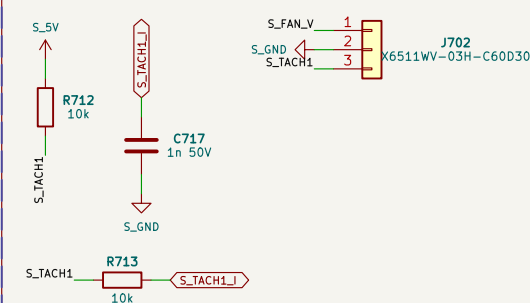
CAN and RS-485 bus sides, SELV side of the three digital isolators; EN floating = on (internal pull-up). 22 uH, 2 x 22 uF.

FAN SUPPLY (TPS54360B, 300 KHZ): SPEED BY VOLTAGE, PASSES WHAT THE WINDING GIVES



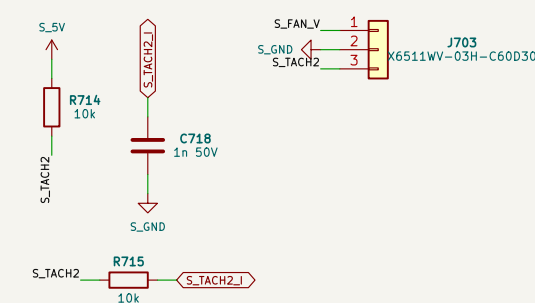
S_FAN_PWM (isolated, 0/5 V) -> 4.7k / 2.2 uF -> S_FAN_VF: EN via 10k (off below 1.1-1.3 V: duty below -25 %, static low or a dead live side = fans OFF) and current injection into FB through 15.0k: higher duty = lower voltage (24 V ... 7 V); at a low input it runs in dropout. Rtop 100k / Rbot 2.87k.

FAN 1



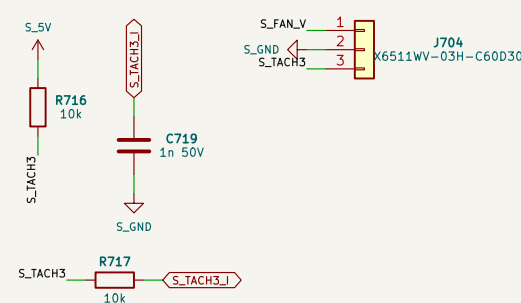
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator input.

FAN 2



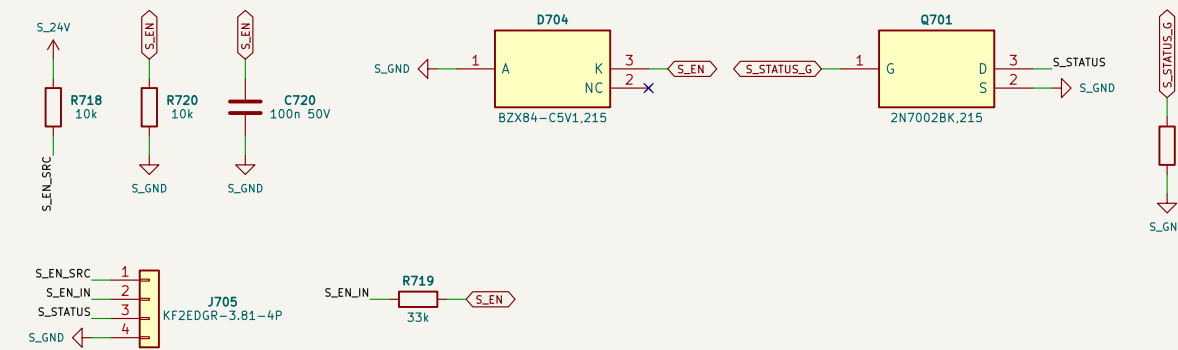
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator Input.

FAN 3



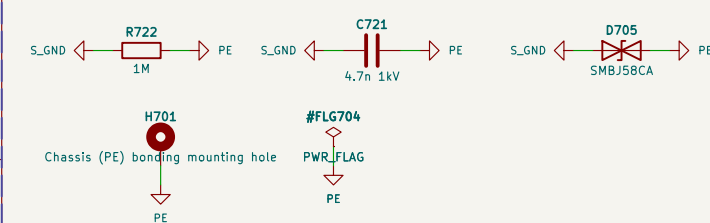
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator input.

ENABLE (EXTERNAL STOP) AND STATUS TERMINAL



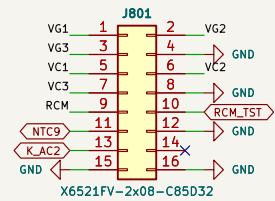
1 S_EN_SRC (S_24V via 10k; dry-contact supply), 2 S_EN_IN (contact return or a PLC output), 3 STATUS (open drain 2N7002BK, conducting = healthy), 4 0 V. S_EN_IN $\rightarrow$ 33k / 10k (5.1 V clamp, 100 nF: 0.77 ms)

SELV 0 V TO PE (SELV MAY BE EARTHED BY THE INSTALLER: PELV)



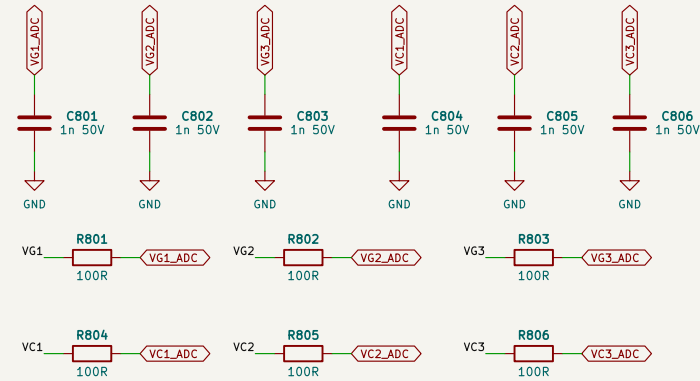
1 M || 4.7 nF 1 kV plus SMBJ58CA: a common-mode surge on the field cables is clamped to PE; a solid PELV bond by the installer is harmless. Plated M3 hole = PE / shield bond near the terminals.

PCS_X CONNECTOR (GEN/INTERFACES.PY PCS_X; AGND = GND ON THIS BOARD)



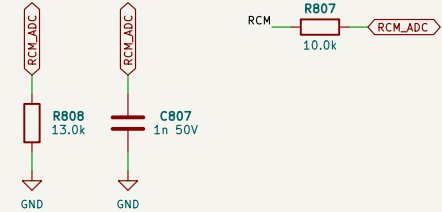
VGN (four-wire build) not connected (OPEN 4), K_AC2 = K_AC2_M AND HEALTHY: a trip opens AC contactor 2 as AC contactor 1. RCM_TST straight from GPIO21 (100k pull-down on PCS-PWR; open = no test current).

GRID AND C_F VOLTAGES (VMID + V/1203 FROM OPA2388 FOLLOWERS ON PCS-PWR)



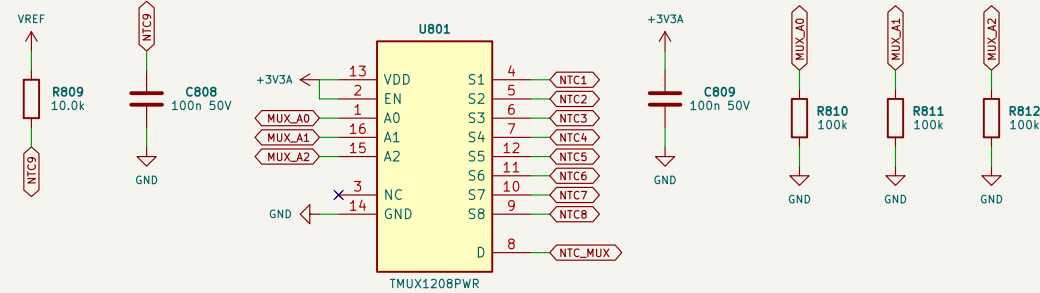
100 R / 1 nF COG charge buckets as VA / VB. VG1-3 on ADC-A/B/C and VC1-3 on ADC-A/B/C: each set sampled together (synchronisation, relay test VG against VC with one contactor closed).

RESIDUAL CURRENT (TYPE-B SENSOR, RFQ ON PCS-PWR; OUTPUT LEVEL ASSUMED, OPEN 5)



10.0k / 13.0k: a sensor output up to its 5.25 V supply reads <= 3.0 V; 1 nF COG charge bucket.

NTC9 (DC-LINK BANK) AND THE NTC MUX



NTC9: Murata NCP15 on PCS-PWR, 10.0k from VREF + 100 nF here. TMUX1208: S1-S8 = NTC1-8 (each keeps its 100 nF and its hardware OT / open-probe comparators, sheet 04) -> NTC_MUX, one ADC pin: A0-A2 from GPIO58-60 with 100k pull-downs. EN high. Firmware converts >= 5 ms after an address change.

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 9
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /08 PCS_X: grid voltages, RCM, NTC9, K_AC2/
File: 08_pcsx.kicad_sch

Title: PCS-CTL control board - PCS_X: grid voltages, RCM, NTC9, K_AC2

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